

Design for class “InvoiceScreen”

Inheritance of class “CustomerScreen”

InvoiceScreen
- placeOrderController : PlaceOrderController - currentOrder : Order
+ setBController(bController : BaseController) : void + navigateTo(screen : BaseScreenHandler) : void + setController(controller : PlaceOrderController) : void + setOrderDetails(order : Order) : void + handleConfirmPaymentClick() : void + displayInvoice(order : Order) : void

Table 1. Attribute Design

#	Name	Data type	Default value	Description
1	placeOrderController	PlaceOrderController	null	Controller for finalizing the order.
2	currentOrder	Order	null	The order waiting for user confirmation.

Table 2. Operation Design

#	Name	Return type	Description
1	setController(controller)	void	Assign the controller for order confirmation.
2	setOrderDetails(order)	void	Display subtotal, shipping fee, and total amount.
3	handleConfirmPaymentClick()	void	Proceed to payment request.
4	display(invoice)	void	Render invoice information on UI.

Method Design:

1. setController

- Parameter:
 - controller: PlaceOrderController - controller that will handle invoice actions.
- Exception:
- Method:
 - Assigns the controller for order confirmation.

2. setOrderDetails

- Parameter:
 - order: Order — the order whose invoice details are to be displayed.
- Exception:
 - NullPointerException if order is null (caller should avoid).
- Method:
 - Read order.subtotal, order.deliveryInfo.fee, order.vat, compute total.
 - Populate UI labels: subtotal, shipping fee, VAT, total.
 - Keep a reference to currentOrder = order for later confirmation action.

3. handleConfirmPaymentClick

- Parameter:
- Exception:
 - PaymentInitiationException if payment subsystem cannot be reached or parameters invalid.
- Method:
 - Validate currentOrder is present and in correct status.
 - Ask the user to select a payment method (e.g., QR or PayPal).
 - Call placeOrderController.requestPayment(method) with chosen method.
 - If controller reports error, show error via AlertDialog.show(message).

4. display

- Parameter:
 - invoice: Object or invoice: Order (the class using it should pass the rendered invoice object)

- Exception:

- Method:

- Render the provided invoice object on screen (refresh UI components).
- Ensure totals and breakdown are consistent with currentOrder.
- Support re-layout or print-preview if available.